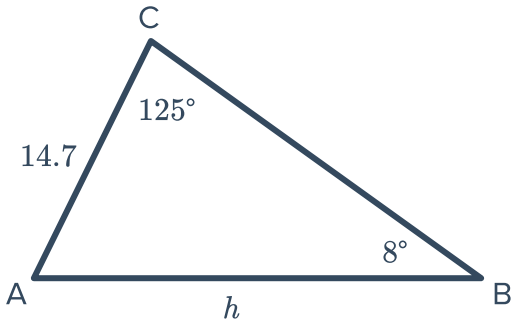
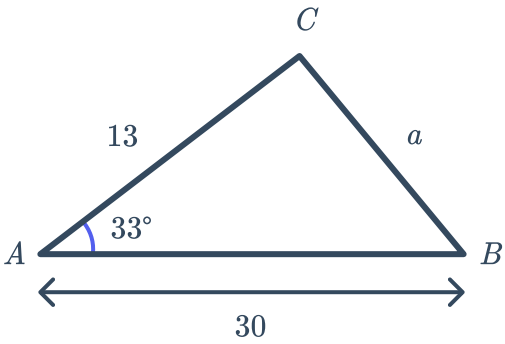
Mixed applications

1 Find the value of the pronumeral in each of the following diagrams. Round your answers to two decimal places.

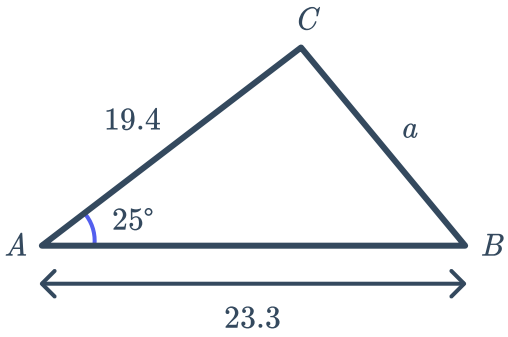
a



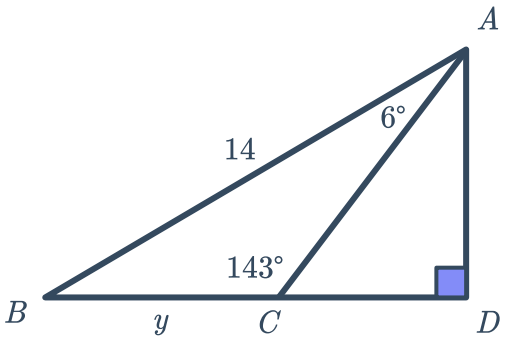
b



c

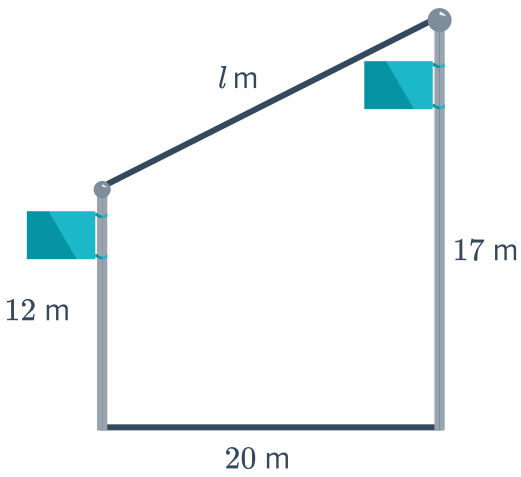


d



2 Two flag posts of height 12 m and 17 m are erected 20 m apart.

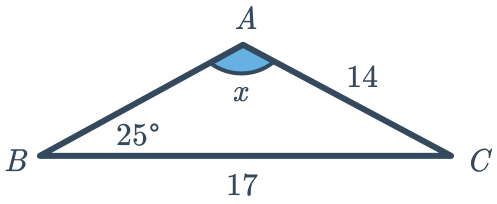
Find the length, l, of the string needed to join the tops of the two posts. Round your answer to one decimal place.



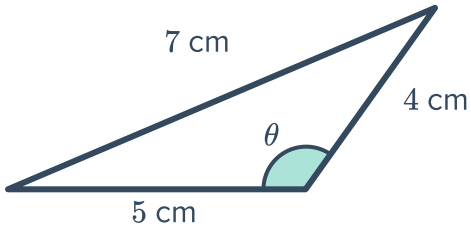
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3 Find the value of the pronumeral in each of the following diagrams. Round your answers to two decimal places.

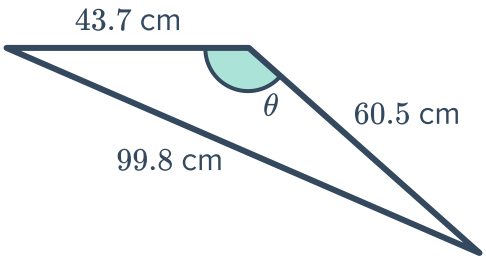
a



b



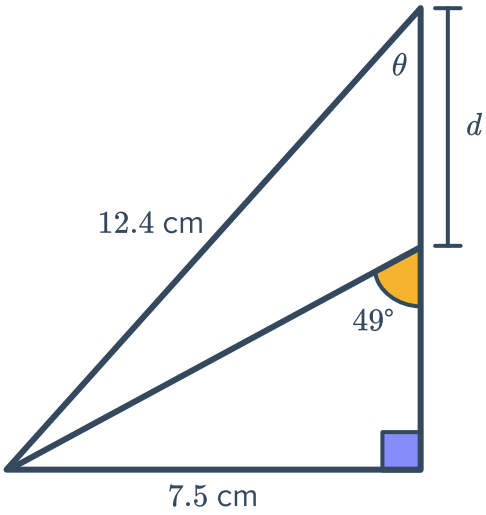
c



4 Consider the following diagram:

a Find the value of θ. Round your answer to two decimal places.

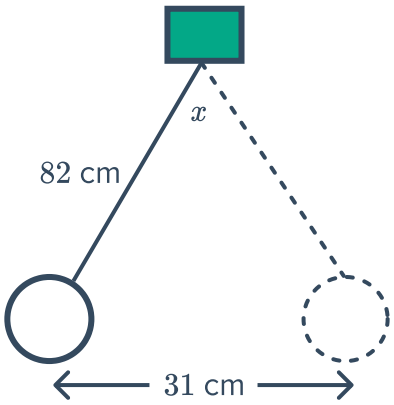
b Hence or otherwise, find the value of d. Round your answer to one decimal place.



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5 A pendulum of length $82\ \text{cm}$ swings a horizontal distance of $31\ \text{cm}$.

Find the angle x of the pendulum's movement. Round your answer to the nearest degree.



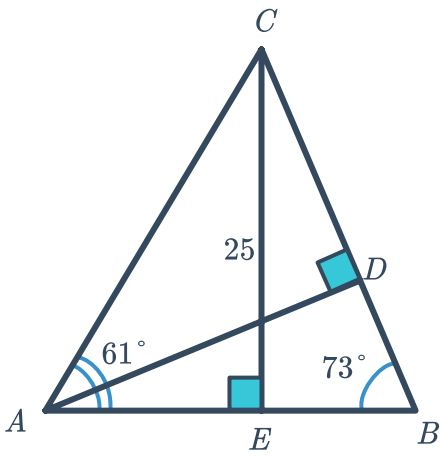
6 In the following diagram, $\angle CAE = 61^\circ$, $\angle CBE = 73^\circ$ and $CE = 25$.

a Find the length of AE, correct to four decimal places.

b Find the length of BE, correct to four decimal places.

c Hence, find the length of AB, correct to two decimal places.

d Find the length of BD, correct to one decimal place.

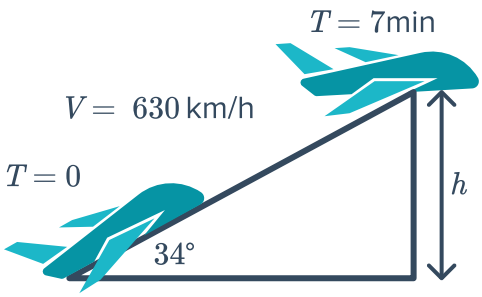


7 A jet takes off and leaves the runway at an angle of 34°. It continues to fly in this direction for $7\ \text{min}$ at a speed of $630\ \text{km/h}$ before levelling out.

a Find the distance in metres covered by the jet just before levelling out.

b If the height of the jet just before levelling out is $h\ \text{m}$, calculate h.

Round your answer to the nearest metre.

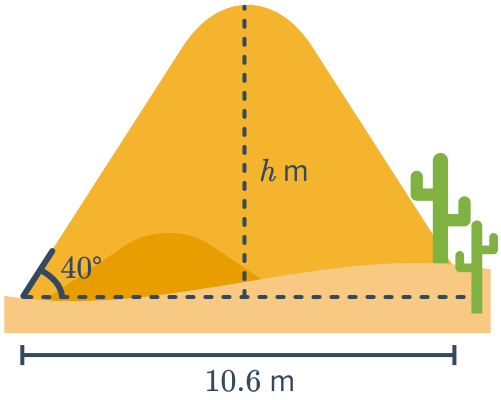


8 A sand pile has an angle of 40° and is

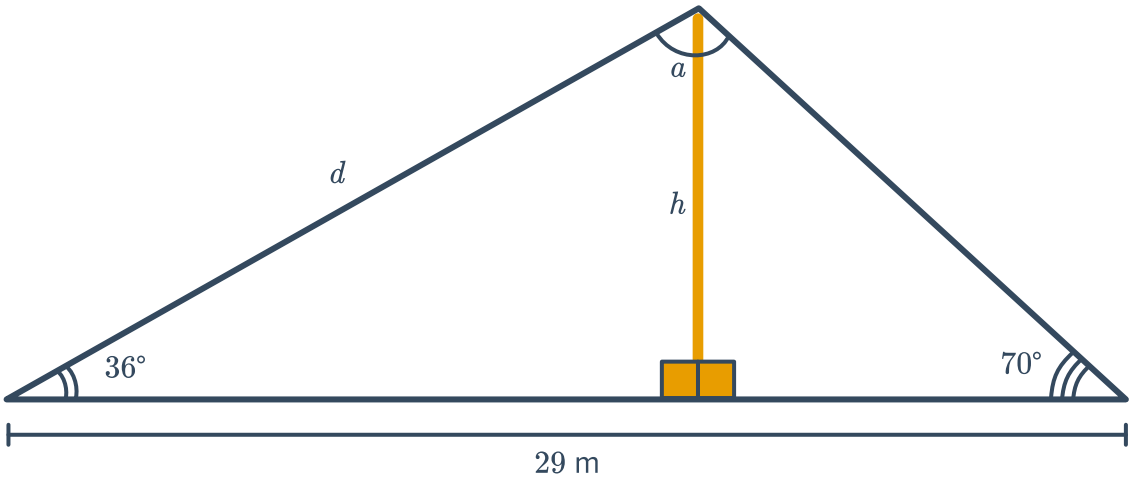
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10.6 m wide.

Find the height of the sand pile, h, to one decimal place.



9 Two wires help support a tall pole. One wire forms an angle of 36° with the ground and the other wire forms an angle of 70° with the ground. The wires are 29 m apart:



a Find a, the angle made between the two wires at the top of the pole.

b Find d, the length of the longest wire in metres. Round your answer to two decimal places.

c Find calculate h, the height of the pole in metres. Round your answer to two decimal places.

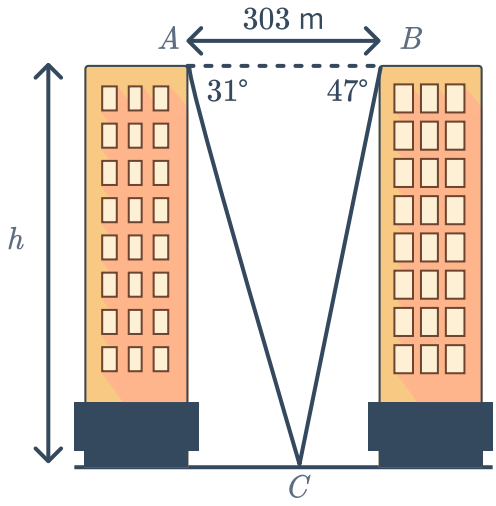
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10 To calculate the height of each block of flats, a surveyor measures the angles of depression from A and B to C. From A, the angle of depression is 31° and from B the angle of depression is 47°.

a Find the size of $\angle ACB$.

b If the distance between A and C is b m, find the value of b. Round your answer to two decimal places.

c If the buildings are h m tall, find the value of h. Round your answer to the nearest metre.

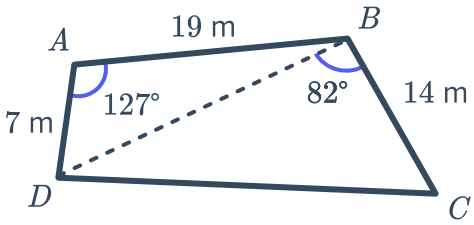


11 A garden, in the shape of a quadrilateral, is represented in the following diagram: Find the following, rounding your answers to two decimal places:

a The length of BD.

b The length of CD.

c The perimeter of the garden.

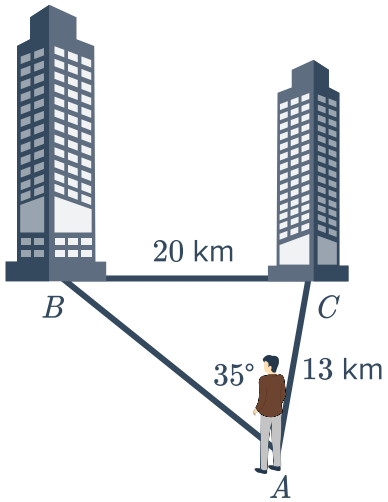


12 Dave is standing on a hill and can see two buildings in the distance. The buildings are 20 km apart.

Dave is 13 km from one building and the angle between the two lines of sight to the buildings is 35°.

a Find the size of $\angle ABC$. Round your answer to two decimal places.

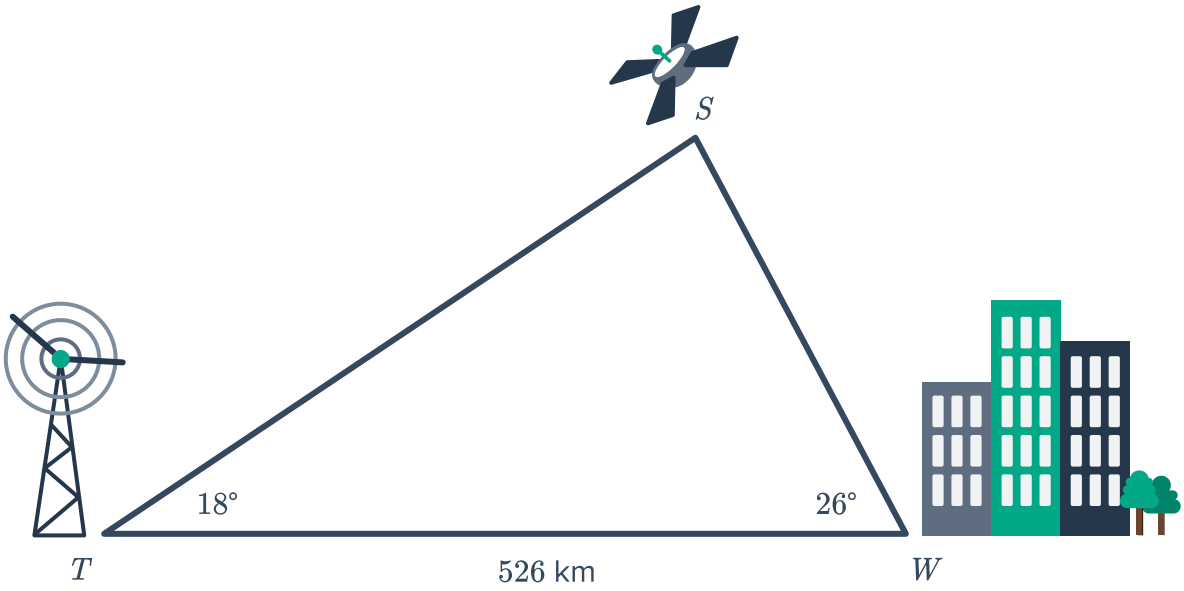
b Find the size of $\angle BCA$. Round your answer to two decimal places.



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c Find the distance between Dave and the building at B, AB. Round your answer to one decimal place.

13 A radio signal is sent from a transmitter at tower T, via a satellite S, to a town W, as shown in the diagram below. The town is $526\ \text{km}$ from the transmitter tower. The signal is sent out from the transmitter tower at an angle of 18°, and the town receives the signal at an angle of 26°.



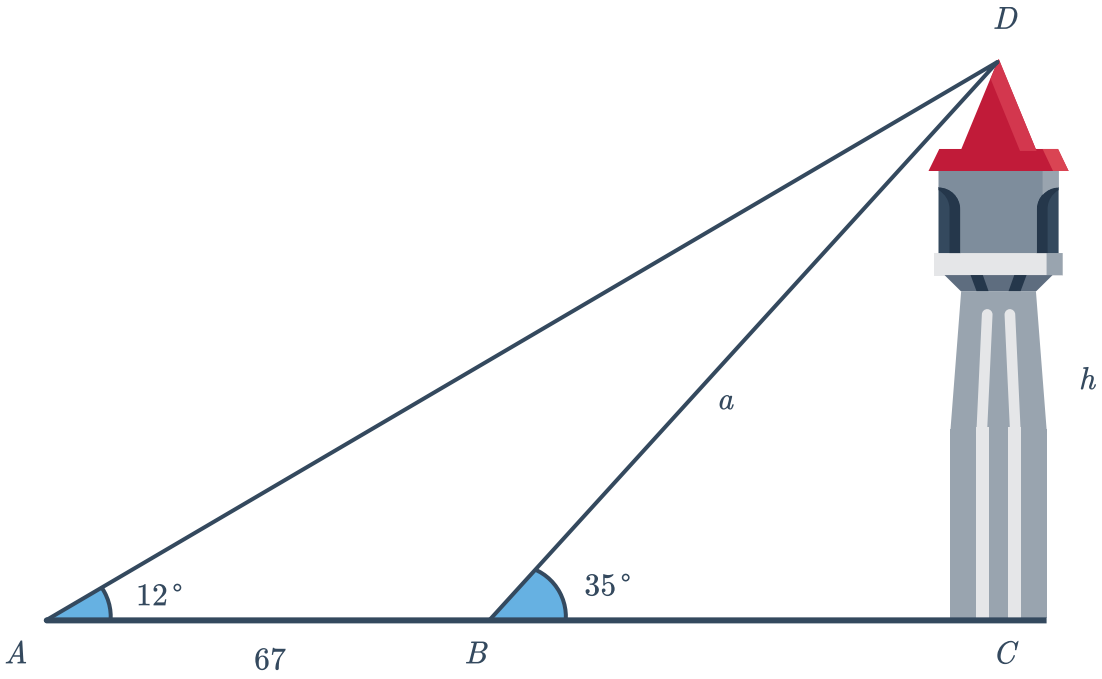
a Find the size of $\angle WST$.

b The signal travels along SW from the satellite to the town. Find the distance it travels, SW, to the nearest kilometre.

c If the satellite is h kilometres above the ground, find h. Round your answer to two decimal places.

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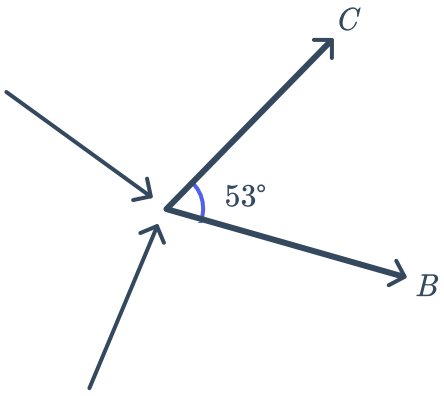
14 Mae observes a tower at an angle of elevation of 12°. The tower is perpendicular to the ground. Walking $67\ \text{m}$ towards the tower, she finds that the angle of elevation increases to 35°.



- **a** Calculate the angle $\angle ADB$.
- **b** Find the length of the side a. Round your answer to two decimal places.
- **c** Hence, find the height h, of the tower. Round your answer to one decimal place.

15 After two meteoroids collide at point A, one starts travelling in the direction of point B, while the other starts travelling in the direction of point C, with an angle of 53° between the two directions. The meteoroid projected in the direction of B is moving at a speed of $7860\ \text{km/h}$, while the other is moving at a speed of $10\,170\ \text{km/h}$.

- **a** What distance will the meteoroid travelling towards point B have covered after 29 minutes after the collision?
- **b** What distance will the meteoroid travelling towards point C have covered after 29 minutes after the collision?
- **c** Find the distance between the two meteoroids $29\ \text{min}$ after the collision. Round your answer to the nearest tenth of a kilometre.

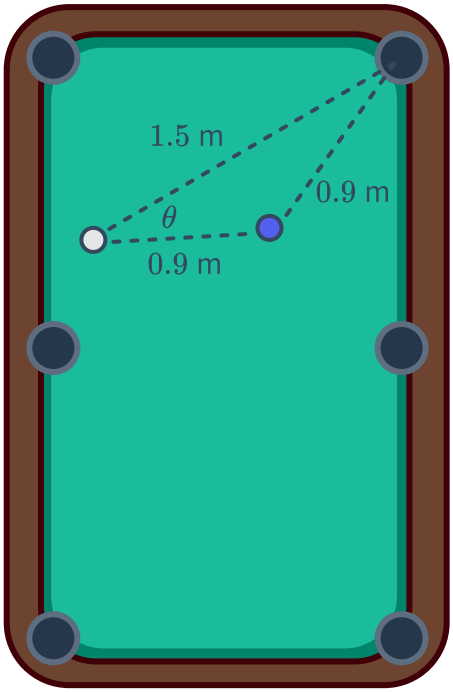


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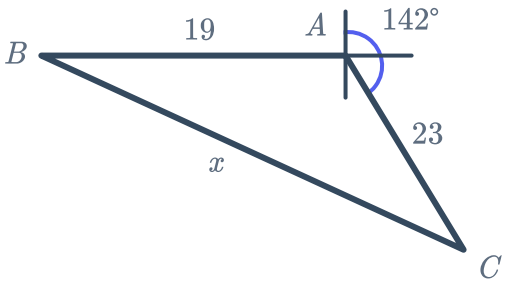
16 In a game of pool, a player has one last ball to sink into a corner pocket. The player must use his cue (stick) to hit the white ball so that it knocks the purple ball into the corner pocket.

The player judges that the white ball is about $1.5\ \text{m}$ away from the corner pocket, and that the distance between the two balls is about $0.9\ \text{m}$, while the purple ball is also $0.9\ \text{m}$ from the corner pocket.

He wants to find the angle θ at which he needs to knock the white ball against the purple ball. Round your answer to two decimal places.



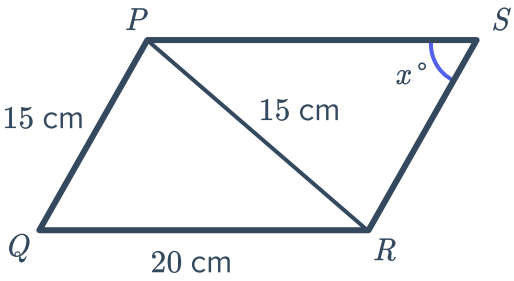
17 Point C has a bearing of 142° from Point A. If Point B is $19\ \text{km}$ West of Point A, determine the distance, x, between Point B and Point C.



18 Consider the given parallelogram:

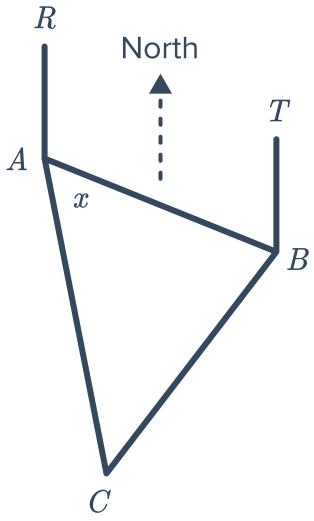
a Find the value of x. Round your answer to the nearest degree.

b Hence, find the size of $\angle SRQ$. Round your answer to the nearest degree.



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19 Christa and James set off for a walk. They leave Point A and walk on bearing of 101° for $4\ \text{km}$ to Point B. Christa then stops to rest but James continues walking on a bearing of 191° for $2\ \text{km}$ to Point C.



a Find $\angle ABC$.

b Find x, to the nearest degree.

c Hence, find the true bearing of A from C, to the nearest degree.

20 The Australian 50 cent coin has the shape of a dodecagon (it has 12 sides). Eight of these 50 cent coins will fit exactly on an Australian $\$10$ note that is $x\ \text{cm}$ tall.

a Find the total area of the eight coins in terms of x.

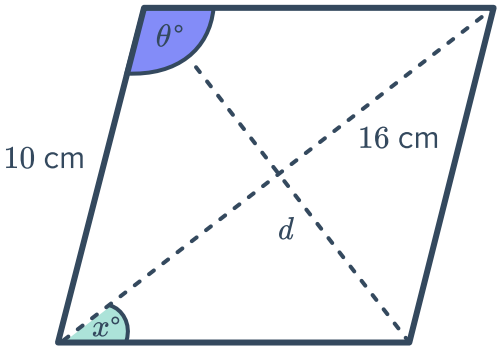
b Find the fraction of the $\$10$ note that is not covered.



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21 A rhombus of side length 10 cm has a longer diagonal of length 16 cm.

Find the following, rounding your answers to one decimal place:

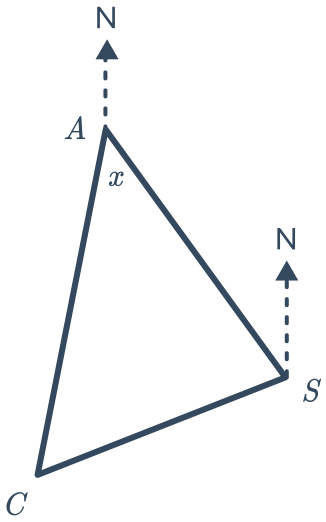


a θ, the obtuse angle.

b x

c d, the length of the shorter diagonal.

22 A commercial passenger plane flies 1801 km on a bearing of 339° from Sydney (S) to Albury (A). A second smaller plane leaves Sydney on a bearing of 249° and loses radio contact at location C after flying for 1301 km .



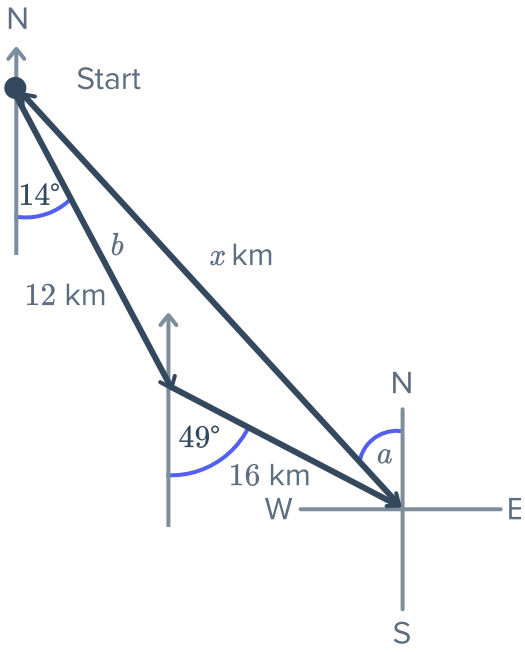
a Find the size of $\angle ASC$.

b Find AC, the distance the passenger plane must fly to reach point C, to the nearest km.

c Find the value of x to the nearest degree.

d Find the true bearing that the passenger plane must fly from point A to reach the smaller plane at point C.

23 A boat travels $\text{S } 14^\circ\text{E}$ for 12 km and then changes direction to $\text{S } 49^\circ\text{E}$ for another 16 km.



a Find x, the distance of the boat from its starting point to two decimal places.

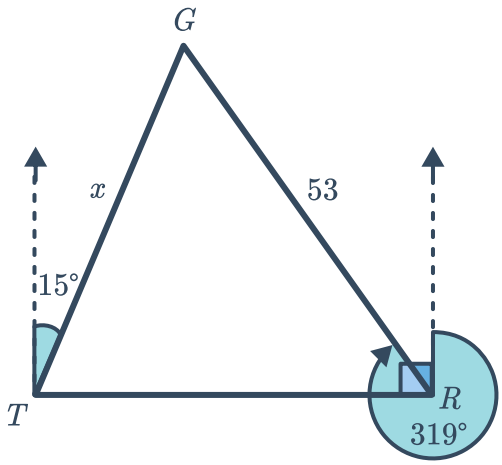
b Find b to the nearest degree.

c Hence, find the bearing that the boat should travel on to return to the starting point.

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24 Grenada (G), Tangiers (T) and Roma (R) are three towns. Grenada bears 15° from Tangiers and 319° from Roma. Tangiers is due west of Roma. The distance from Grenada to Roma is 53 km.

Find the distance from Grenada to Tangiers, x, to the nearest kilometre.



25 Find the exact side length, y, of an equilateral triangle with a perpendicular height of $\sqrt{21}$ cm.

26 The angle of elevation to the top of a 33-metre high tower is 35° from point A, due west of the tower. The point B is located 60 metres due south of point A.

a Find the distance, from point A to the base of the tower, correct to two decimal places.

b Find the distance from point B to the base of the tower, correct to one decimal place.

c Find θ, the angle of elevation from point B to the top of the tower, correct to the nearest degree.

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27 A student created a scale model of Australia and drew a triangle between Alice Springs, Brisbane and Adelaide:



a Find the angle θ, between Brisbane and Adelaide from Alice Springs. Round your answer to one decimal place.

b Hence or otherwise, find the area taken up by the triangle. Round your answer to two decimal places.

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